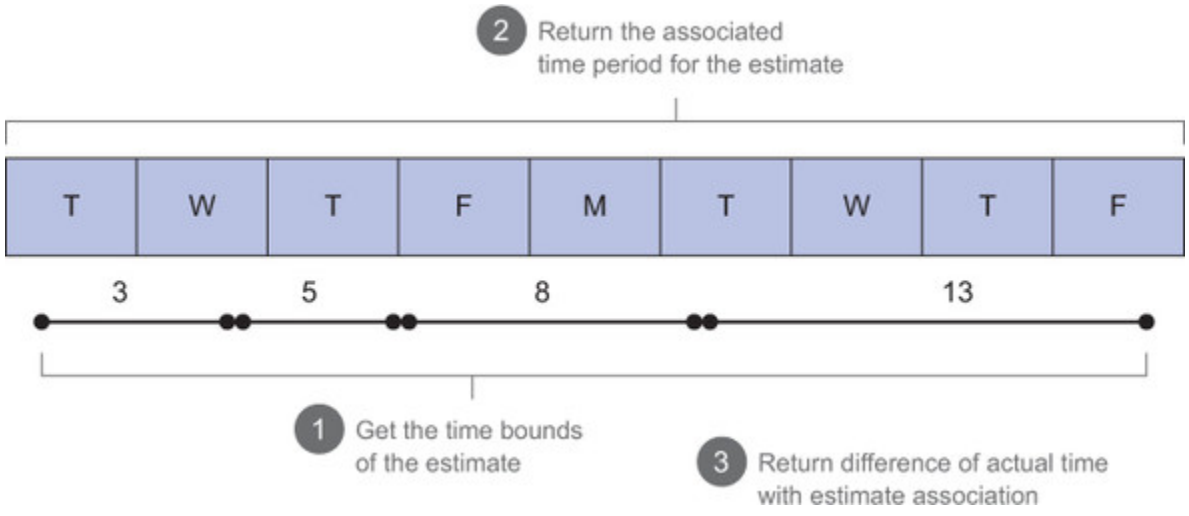
other estimates in your estimation scale to see if the amount of time a task took matches the corresponding time window for the estimate.

To map estimates to time you'll need a function that can find the time bounds you're estimating within, the time scale, and the estimate scale. Figure 7.14, shows how this can work as a function.

Figure 7.14. A visual representation of the algorithm to determine estimate health



An algorithm to do this is outlined in the following listing.

Listing 7.1. Algorithm for checking estimate health

```
static def estimateHealth(estimate, actualTime, maxEstimate, maxTime,
    estimationValues) {
  def result
  def timeEstimateRatio =  maxTime / maxEstimate
  def estimateTime = estimate * timeEstimateRatio
  def upperTimeBound = maxTime
  def lowerTimeBound = 0

  def currentEstimateIndex = estimationValues.findIndexOf { it ==
  ➥ estimate}

  if(currentEstimateIndex == 0) {
    lowerTimeBound = 0
  } else {
    lowerTimeBound = estimateTime - ((estimateTime -
     ➥ (estimationValues[estimationValues.findIndexOf { it == estimate} - 1]
     ➥ * timeEstimateRatio)) / 2)
    }
```

Initializes the variables

Finds the input's index in the array

Lower bound of 0 for the lowest estimate

Calculates the lower time bound